

Wave 5.2R Integrated-Specialist Track 2 Official Decision

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Decision

The Wave 5.2R integrated-specialist refresh completed separate `Fw` , `Bw` , and direction-aware `global` matrices and passed the required curve-first review. The official outcome remains `forward_harmonic_specialist_added` :

- recommend `wave52r_integrated_a02_seed_314159` as the verified offline `Fw` specialist and as the best routed `global` integrated candidate;
- retain `wave52r_promotion_k01_global_seed_271828` as the `Bw` recommendation and frozen temporal backbone;
- retain A03-A07 as diagnostic ablations because each failed its campaign branch gate, even when it wins an isolated Track 2 axis;
- keep all accepted periodic GRU and periodic harmonic MLP registries unchanged;
- make no TwinCAT runtime or deployment-readiness claim.

A02 is not a new backward model. Its backward prediction is effectively the frozen K01 path, while its measurable benefit comes from the deterministic forward-gated centered H08 contribution.

Evidence Scope

The operator-run matrix evaluated the prepared 29-candidate inventory with surface filtering:

Surface	Curves	Candidates	New trained runs	Control role
<code>Fw</code>	97	27	21	K01 global, H08 Fw, and four compatible incumbent surfaces
<code>Bw</code>	97	26	21	K01 global and four compatible incumbent surfaces
<code>global</code>	194	24	21	K01 global and two global incumbent surfaces

The CVP 1.2 shortlist contains the candidates that won at least one raw or robustness axis, the only campaign-qualified branch, and the frozen K01/H08 controls. A08 is omitted as an independent shortlist entry because it contains only A02 and is prediction-equivalent to it.

The visual shortlist adds the six accepted incumbent surfaces, for 15 plotted candidates. The official comparison therefore preserves incumbent evidence without treating a repeated full-payload diagnostic as a prerequisite for their already established roles.

Selection Method

The decision applies the canonical multi-index curve-first policy with bounded within-surface rank normalization. Lower scores are better.

Block	Weight	Metrics used
Shape and harmonic fidelity	35%	centered MAE, peak-to-peak error, harmonic amplitude error, harmonic phase error
Raw operating error	20%	curve MAE, RMSE, mean percentage error
Offset and continuity	20%	absolute mean-offset error, closure mismatch
Robustness	15%	P95 mean percentage error
Deployment readiness	10%	causal/export evidence, added complexity, inspectability, campaign gate status

Campaign branch-gate failure is a promotion veto. The frozen H08 candidate also keeps its earlier raw/offset regression veto. These candidates remain visible as axis winners but cannot become the final recommendation.

Per-Surface Winners

Surface	Best raw	Best shape	Best offset	Best P95	Recommendation
Fw	A04 seed 161803	A02 seed 314159	A07 seed 161803	A07 seed 314159	A02 seed 314159
Bw	A03 seed 161803	A03 seed 161803	A07 seed 161803	A07 seed 161803	K01 global seed 271828
global	A04 seed 161803	A04 seed 161803	A07 seed 161803	A06 seed 271828	A02 seed 314159

A03, A04, A06, and A07 win diagnostic axes but remain vetoed by their failed campaign gates. This is exactly the distinction required by the policy: an isolated metric win does not override failed specialty or non-regression evidence.

Forward Decision

A02 seed 314159 improves over frozen global K01 on the forward matrix:

- raw MAE decreases from 0.001404833 deg to 0.001400449 deg , a 0.312% improvement;
- P95 mean percentage error decreases by 0.669% ;
- centered-shape MAE decreases from 0.001153412 deg to 0.001147969 deg ;
- mean harmonic-amplitude error decreases from 5.095826% to 4.750917% .

K01 remains slightly better on mean peak-to-peak error and mean harmonic phase error. A02 nevertheless has the best veto-free multi-index score and is the only added branch that passed the campaign gate for all three seeds. A04 seed 161803 has the best forward raw MAE and centered-shape MAE, but its campaign specialty and multi-index non-regression gates failed. A07 wins the forward robustness axis but also failed its campaign gate. Neither is promoted. The forward recommendation is therefore A02 seed 314159 as a verified offline forward specialist. This does not yet replace an accepted deployment registry.

Backward Decision

A02 and frozen K01 are effectively tied on Bw :

- A02 raw MAE is 0.001523088 deg ;
- K01 raw MAE is 0.001523086 deg ;
- their P95 mean percentage errors differ by less than 0.004% relative.

This equivalence matches the deterministic design: the H08 contribution is zero on backward curves. A03 seed 161803 improves raw and centered-shape metrics, while A07 improves offset and P95 behavior, but both branches failed their campaign gates. The backward recommendation remains K01 global seed 271828 . There is no distinct integrated-specialist backward promotion.

Global Decision

A02 seed 314159 improves global raw MAE over K01 from 0.001463960 deg to 0.001461769 deg , a 0.150% improvement. Its CVP 1.2 diagnostic score is 3.652449 , the best of the nine-candidate payload shortlist. The global gain is the aggregation of a real forward improvement and an unchanged backward path; it is not evidence for a new undifferentiated global expert.

A04 seed 161803 is the global raw and shape winner, and A06/A07 win robustness and offset axes, but all retain campaign-gate vetoes. A02 is therefore the best veto-free routed global integrated candidate. K01 remains the frozen temporal backbone and the simpler backward recommendation.

Visual Review

The reviewed A02 collage and forward/backward overlays show:

- continuous full-revolution predictions with no visible stitching failure;
- close agreement among the integrated variants on ordinary conditions;
- under-reproduction of high-frequency amplitude at the hardest high-speed, high-torque examples across both new and incumbent groups;
- a measurable but visually small forward benefit from the specialist branch;
- no distinct backward A02 behavior beyond K01-equivalent routing.

The plots therefore support the numerical recommendation but also show why the offline improvement must not be described as a solved compensation problem. The extreme-condition amplitude gap remains visible.

Incumbent And Deployment Boundary

A02 raw MAE is lower than the accepted directional or global incumbent models on every applicable matrix. Relative to the matching accepted directional incumbents, its raw-MAE improvements are approximately:

- 13.47% versus periodic GRU Fw and 17.31% versus harmonic MLP Fw;
- 17.09% versus periodic GRU Bw and 20.35% versus harmonic MLP Bw;
- 19.25% versus periodic GRU global and 15.69% versus harmonic MLP global.

These are offline comparisons only. The accepted incumbents retain their registry and deployment roles because A02 has not passed the separate export, TwinCAT build, target activation, ADS, license, and commissioned runtime gates. Static or host-side parity cannot substitute for those checks.

Final Status

The official Track 2 decision is

- `FW` : A02 seed `314159` is a verified offline forward-specialist candidate;
- `Bw` : retain K01 global seed `271828` ;
- `global` : A02 seed `314159` is the preferred routed integrated offline candidate, with benefit attributable only to its forward branch;
- A03-A07: retain as diagnostic failed-gate ablations;
- A08: retain as the prediction-equivalent A02 composition replay;
- accepted registries: unchanged;
- deployment readiness: not established.

The next model-facing step is export and TwinCAT qualification of the selected A02 path if the project chooses to advance it. Manual standalone PLC testing may continue in parallel without changing this offline decision. The live backlog, training master summary, program closeout ledger, active campaign state, and canonical directional overview were synchronized. The RCIM Paper Reference Benchmark was checked and requires no content change because this refresh contains no repository-owned online compensation result.

Artifact Index

- Matrix summaries:
`output/validation_checks/track2_reference_comparison/2026-08-03-21-58-09__wave52r_integrated_specialist_track2_wave52r_integrated_specialist_track2_forward/validation_summary.yaml` , `output/validation_checks/track2_reference_comparison/2026-08-03-21-59-38__wave52r_integrated_specialist_track2_wave52r_integrated_specialist_track2_backward/validation_summary.yaml` , and `output/validation_checks/track2_reference_comparison/2026-08-03-22-01-08__wave52r_integrated_specialist_track2_wave52r_integrated_specialist_track2_global/validation_summary.yaml` .
- CVP 1.2 diagnostics:
`output/validation_checks/wave52r_integrated_specialist_track2_curve_payload_diagnostics/2026-08-04-00-31-50__track2c_curve_payload_diagnostics/` .
- Multi-index decision:
`output/analysis/wave_5_2r/integrated_specialist_track2_decision/multi_index_surface_decision.yaml` .
- Candidate scores:
`output/analysis/wave_5_2r/integrated_specialist_track2_decision/multi_index_candidate_scores.csv` .
- Collage report:
`doc/reports/analysis/te_curve_verification_pipeline/02_visual_reports/wave52r_integrated_specialist_best_model_collage_report/[2026-08-04]/track2_best_model_collage_report.md` .
- Overlay report:
`doc/reports/analysis/te_curve_verification_pipeline/02_visual_reports/wave52r_integrated_specialist_multi_model_curve_comparison_report/[2026-08-04]/track2_multi_model_curve_comparison_report.md` .